

FIELD GUIDE TO TREES

Grow a tree.

Learn how it holds together.

Insert values into a Binary Search Tree, AVL Tree, Red-Black Tree, or Trie, and watch the structure rebalance in real time. Then switch to the challenge card and predict traversal order to earn growth rings.

- Binary Search Tree
- AVL Tree
- Red-Black Tree
- Trie

BINARY SEARCH TREE – INSERT NUMBERS, NO REBALANCING

7 NODES

node

```
graph TD; 50((50)) --- 30((30)); 50 --- 70((70)); 30 --- 20((20)); 30 --- 40((40)); 70 --- 60((60)); 70 --- 80((80));
```

HEIGHT

NODE COUNT

MAX |BALANCE|

3

7

0

CONTROLS

Insert

Random value

Clear tree

TRAVERSAL CHALLENGE

streak 0

Sapling

0 pts

Insert at least 3 nodes to unlock a challenge.

New challenge

Undo last pick



Drag isn't needed here – insert values, click nodes in the challenge to answer. Layout redraws on every insertion.